Inorganic CUME on Materials Chemistry
December 2nd, 2016
Answer all the following questions

1. (20 points) Write down the reaction for the lithiation of Silicon (Si) i.e., Li-Si alloy reaction; What are the structural differences between crystalline and amorphous Si? What are the similarities and differences between crystalline and amorphous Si during the electrochemical lithiation and de-lithiation process according to this paper?

2. (20 points) Briefly discuss what is Transmission Electron Microscopy (TEM)? How is it usually used for materials characterization? What are the differences between *ex situ* and *in situ* TEM? Why the authors need to use a thin layer of Li_2O to coat the Li surface when designing the *in situ* TEM cell in the paper (see Figure 1a in the paper)?

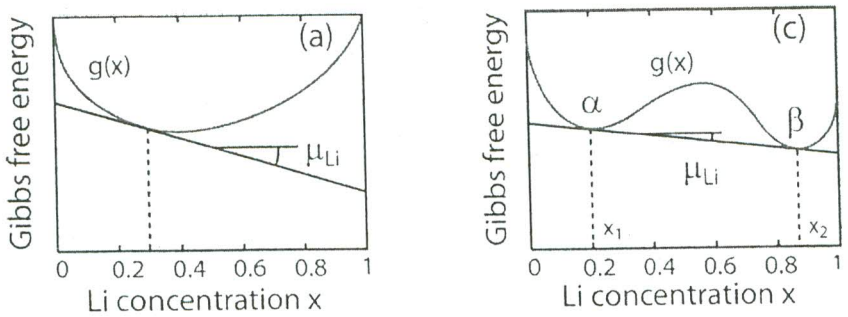
3. (20 points) What factors affect the rate of diffusion in general? Instead of a diffusion-controlled single-phase process, the authors claim that the lithiation of amorphous Si in this paper is a two-phase lithiation process. Explain why in your own word (up to 5 sentences) based on the information provided in this paper. Name a few other techniques that can be used to differentiate solid solution and two phase reactions and briefly discuss why.

4. (20 points) Explain how the tension forms at the surface of crystalline Si during the two-phase lithiation process. Why there is no fracture observed in amorphous Si sphere during lithiation? What are the main challenges existing in employing Si as the anode for next-generation batteries? Why does amorphous Si help? How to take advantage of nanotechnology to address the challenges associated with Si anodes?

5. (20 points) The chemical potential of interstitial Li in an electrode material is equal to the derivative of the free energy of the material with respect to Li concentration according to

$$\mu_{Li} = \frac{\partial g}{\partial x}$$

Here g is the Gibbs free energy per LixM formula unit, where M represents the host chemistry (e.g., Si, CoO_2 etc) and x denotes the fraction of available sites occupied by Li. Graphically, the Li chemical potential within a compound at a particular composition x is equal to the slope of the free energy g at that composition as illustrated in the Figures 1a-b below. The straightforward connection between a measurable voltage curve and the Gibbs free energy means that the occurrence and nature of phase transformations due to variations in Li concentration will have clear signatures in the voltage profile. For example, if Si forms a solid solution with Li as in Li_xSi during lithiation process (see Figure 1a below, assume Li concentration is up to 1 during the reaction), it will exhibit a smooth sloping voltage curve as illustrated in Figure 1b below. If the lithiation process of Si is accompanied by a first-order phase transformation from a Li poor phase, α, to a Li rich phase, β (Figure 1c), predict the corresponding voltage curve and explain why.



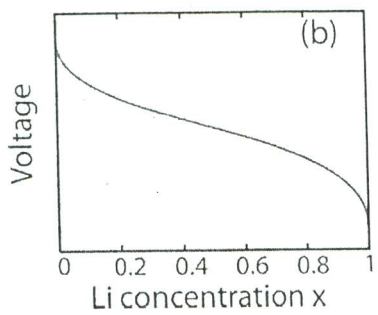


Figure 1. (a)The Gibbs free energy and (b) the voltage profile of a Li-hosting materials undergoing a solid solution reaction with Li; (c) the Gibbs free energy of a Li-hosting compound exhibiting a first-order phase transition upon lithiation. Voltage curves are linearly related to the slope of the free energy of the electrode material.